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Information processing device and method

Abstract

According to embodiments, a device includes a camera configured to read a symbol attached to a commodity and capture an image containing discount information attached to the commodity; a commodity information storage configured to store information regarding the commodity acquired by reading the symbol attached to the commodity; a first evaluator configured to determine whether prior commodity information stored in the commodity information storage includes information related to invisible information, in response to reading discount information attached to the commodity; a second evaluator configured to determine whether commodity identification information included in the discount information is identical to commodity identification information included in the information related to the invisible information, in response to the first evaluator determining that the prior commodity information stored in the commodity storage includes the information related to the invisible information; and an output generator configured to output information related to the discount information.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION
(1) This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2021-122361, filed on Jul. 27, 2021, the entire contents of which are incorporated herein by reference.
FIELD
(2) Embodiments described herein relate generally to an information processing device and a method.
BACKGROUND
(3) In the related art, for example, symbols such as barcodes or two-dimensional codes are printed on commodities, or printed labels are attached to commodities in stores where the commodities are sold. Information processing devices such as point of sales (POS) terminals read symbols attached to commodities and perform commodity registration processes.
(4) Operators who operate POS terminals face surfaces (to which symbols are attached) toward code readers to read the symbols. Therefore, the operators spend time searching for the surfaces to which the symbols are attached.
(5) In recent years, to reduce such time spent searching by operators, technologies have been created for printing electronic watermarks in which invisible information symbols are embedded on commodities. The invisible information is read with code readers. Since the electronic watermarks are invisible to human eyes, the digital watermarks are printed on a plurality of portions on commodities, improving the operability of reading the invisible information by operators. In addition, designability of commodities can be improved.
(6) Incidentally, if prices of commodities are discounted, discount seals are attached to the commodities. If symbols printed in discount seals are read by operators, the commodities can be sold at discount prices with POS terminals.
(7) However, for example, if discount seals are attached to commodities in which electronic watermarks are attached to a plurality of surfaces, code readers read invisible information before reading the discount seals and continuously read the discount seals. Accordingly, double registration may occur, in which a commodity registration process is performed twice on the commodities with POS terminals.

Description

DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a diagram illustrating an example of a commodity according to a first embodiment;
(2) FIG. **2** is a diagram illustrating a connection relation between a code reader and a POS terminal, according to some embodiments;
(3) FIG. **3** is a block diagram illustrating an example of a hardware configuration of the code reader, according to some embodiments;
(4) FIG. **4** is a block diagram illustrating an example of a hardware configuration of the POS terminal, according to some embodiments;
(5) FIG. **5** is a flowchart illustrating an example of a control process of the code reader, according to some embodiments;
(6) FIG. **6** is a functional block diagram illustrating an example of a functional configuration of the POS terminal, according to some embodiments;
(7) FIG. **7** is a flowchart illustrating an example of a control process of the POS terminal, according to some embodiments;

(8) FIG. **8** is a functional block diagram illustrating an example of a functional configuration of a code reader according to a second embodiment; and
(9) FIG. **9** is a flowchart illustrating an example of a control process of the code reader according to the second embodiment.
DETAILED DESCRIPTION
(10) Disclosed herein are systems and methods to prevent double registration of commodities due, for example, to invisible information and discount seals.
(11) In general, according to some embodiments, an information processing device includes a camera configured to read a symbol attached to a commodity and capture an image containing discount information attached to the commodity; a commodity information storage configured to store information regarding the commodity acquired by reading the symbol attached to the commodity; a first evaluator configured to determine whether prior commodity information stored in the commodity information storage includes information related to invisible information, in response to reading discount information attached to the commodity; a second evaluator configured to determine whether commodity identification information included in the discount information is identical to commodity identification information included in the information related to invisible information, in response to the first evaluator determining that the prior commodity information stored in the commodity storage includes the information related to invisible information; and an output generator configured to output information related to the discount information, in response to the second evaluator determining that the commodity identification information included in the discount information is identical to the commodity identification information included in the information related to the invisible information.
First Embodiment
(12) Hereinafter, an information processing device and a method according to a first embodiment will be described with reference to the appended drawings. In the first embodiment, a POS terminal will be described as an example of the information processing device. It should be appreciated that the first embodiment is an example embodiment of the information processing device, and the method, configurations, functions, and the like do not limit to the example embodiment.
(13) First, an electronic watermark will be described. An electronic watermark is a technology for embedding information regarding an image, in the image, in an invisible state (or a state in which information is hard to be seen). Information that looks like nothing is written at a glance (by the human eye) can be extracted using dedicated software. In the case of a commodity, a symbol such as a barcode or a two-dimensional code is embedded as invisible information (e.g., information invisible to a human eye) on a box or a package.
(14) FIG. **1** is a diagram illustrating an example of a commodity according to the first embodiment. As illustrated in FIG. **1**, an electronic watermark D is printed on a plurality of surfaces of a commodity T. Since the electronic watermark D is invisible or hard to be seen with human eyes, design of a package of the commodity T is not interrupted. In FIG. **1**, a symbol is printed as invisible information F. As illustrated in FIG. **1**, the invisible information F is visible (seen with eyes) for description, however, the invisible information F is actually unseen with human eyes. As illustrated in FIG. **1**, the invisible information F is printed on one side surface of the commodity T, however, the invisible information F is actually printed on a plurality of surfaces of the commodity T as the electronic watermark D. A symbol such as a barcode or a two-dimensional code includes a commodity code (commodity identification information) which is a code for identifying the commodity T or information for acquiring the commodity code. In the following description, the invisible information F may include a commodity code.
(15) When discounting a price of the commodity T, a discount seal R is attached to any one side surface of the commodity T. The discount seal R has a discount information description portion M and a symbol portion K. In the discount information description portion M, a pre-tax price, a tax-included price, and information regarding discount (in FIG. **1**, information about “30% discount

from the pre-tax price”) of the commodity T are printed. In the symbol portion K, for example, a symbol such as a barcode is printed. The symbol printed in the symbol portion K includes, among other information, information regarding a commodity code and information regarding the discount. In the following description, the symbol printed in the symbol portion K is assumed to include a commodity code (e.g., “49xxxxxx” illustrated in FIG. **1**) and information regarding the discount (e.g., information about “30%” indicating a discount of 30% as illustrated in FIG. **1**). The information printed in the discount information description portion M and the symbol portion K is visible information and information seen/visible with human eyes. The discount seal R attached to the commodity T overlaps a part of the invisible information F in some cases. However, since the plurality of pieces of invisible information F are printed on the plurality of surfaces of the commodity T, there is no influence/interruption on reading of the invisible information F by a code reader.

(16) Hereinafter, a code reader and a POS terminal will be described. FIG. **2** is a diagram illustrating a connection relation between a code reader **1** and a POS terminal **3**, according to some embodiments. As illustrated in FIG. **2**, the code reader **1** and a POS terminal **3** are connected via a dedicated connection line **5**.

(17) The code reader **1** includes a camera **19** (see FIG. **3**). The camera **19** includes a light-receiving sensor configured with a charge coupled device (CCD) or a complementary metal oxide semiconductor (CMOS). The code reader **1** is, for example, a stationary type scanner or a handy type touch scanner.

(18) The code reader **1** recognizes information regarding the electronic watermark D attached to the commodity T from an image of the commodity T captured by the camera **19**. Then, the code reader **1** reads the invisible information F recorded on the recognized electronic watermark D. The code reader **1** reads the symbol printed on the discount seal R and attached to the commodity T from the image of the commodity T captured by the camera **19**.

(19) The code reader **1** stores a dedicated program A for reading the invisible information F and a dedicated program B for reading a visible symbol. The code reader **1** starts the program A, recognizes the electronic watermark D, and reads the invisible information F. The code reader **1** starts the program B and reads the visible symbol, that is, the symbol printed on the discount seal R. The code reader **1** starts the programs A and B simultaneously (in real-time or near real-time) or alternately/sequentially to read the invisible information F and the symbol printed on the discount seal R.

(20) The code reader **1** decodes the invisible information F read using the program A to generate code information A (e.g., information regarding a commodity) related to the invisible information F. The code information A includes information (information related to the invisible information) indicating that the commodity code and the invisible information F (information regarding an electronic watermark) are read. The information indicating that the invisible information F is read is, for example, information added if the program A reads the information and/or flags information (a code mark) indicating that the information is read. The information indicating that the invisible information F is read may be information indicating the code information A.

(21) The code reader **1** decodes the symbol (discount information) read from the discount seal R using the program B to generate code information B related to the symbol printed on the discount seal R (e.g., information regarding a commodity). The code information B includes the commodity code, the discount information, and information indicating that the discount seal R is read. The information indicating that the discount seal R is read is, for example, information added if the program B reads the information and/or flags information (e.g., a code mark) indicating that the information is read. The information indicating that the discount seal R is read may be information indicating the code information B. The code reader **1** transmits (outputs) the decoded code information A and code information B to the POS terminal **3** via the connection line **5**.

(22) The POS terminal **3** performs a commodity registration process related to the commodity T

based in part on the code information A received (input) from the code reader **1**. The commodity registration process includes a process of calling and displaying commodity information such as a commodity name and/or a price of the commodity T based on the commodity code included in the code information A. The commodity registration process may also include storing the commodity information (which is commodity information A described in more detail below) in a memory (e.g., commodity information storage **331** in FIG. **4**). The commodity information may include a commodity code, a commodity name, and information indicating the reading from the invisible information F (the information related to the invisible information).
(23) The POS terminal **3** performs a commodity registration process related to the commodity T based in part on the code information B received (input) from the code reader **1**. The commodity registration process includes a process of calling and displaying the commodity name, the price, and the like of the commodity T based on the commodity code included in the code information B. The commodity registration process may also include discounting the commodity based on information related to a discount, and performing a process of displaying a discount amount and a price of the commodity T after the discount. The commodity registration process may also include storing commodity information (which is the commodity information B described in more detail below) in a memory (e.g., commodity information storage **331** in FIG. **4**). The commodity information may include a commodity code, a commodity name, a price of the commodity after discount, information related to the discount, and information indicating the reading from the discount seal R.
(24) In traditional/conventional methods, if the commodity T on which the electronic watermark D is printed and the discount seal R is attached is read by the code reader **1**, the invisible information F is first read normally and the symbol printed in the symbol portion K is continuously read. In this case, the code reader **1** transmits the code information A related to the decoded invisible information F and the code information B related to the decoded symbol to the POS terminal **3**. Accordingly, the POS terminal **3** may perform double registration in which a commodity registration process is performed based on each of the received code information A and the received code information B. As will be described in detail below, when receiving and performing the commodity registration process on the code information B, when code information A (e.g., information regarding the commodity T subjected to the commodity registration process immediately previously is information (read from the electronic watermark D) includes the commodity code of the commodity T such that a commodity code included in the code information A is the same as a commodity code included in the code information B, the POS terminal **3** according to the embodiment erases (deletes) the commodity information related to the code information A subjected to the commodity registration process immediately previously from the commodity information section **331**. Therefore, the POS terminal **3** can prevent double registration even if the discount seal R is attached to the commodity T.
(25) The POS terminal **3** performs a settlement process based on the commodity information regarding the commodity T subjected to the commodity registration process. The settlement process is a process of displaying a sum of money or change money (hereinafter referred to as “settlement information”) related to a transaction with a customer based on the commodity information regarding the commodity T subjected to the commodity registration process. The settlement process includes a process of dispensing a receipt in which the commodity information regarding the settled commodity T or the settlement information is printed.
(26) Hereinafter, hardware of the code reader **1** will be described. FIG. **3** is a block diagram illustrating an example of a hardware configuration of the code reader **1**, according to some embodiments. As illustrated in FIG. **3**, the code reader **1** includes a central processing unit (CPU) **11**, a read-only memory (ROM) **12**, a random access memory (RAM) **13**, and a memory section (memory) **14**. The CPU **11** serves as a control entity. The ROM **12** stores various programs. The RAM **13** loads a program and/or various kinds of data. The memory section **14** stores various

programs. The CPU **11**, the ROM **12**, the RAM **13**, and the memory section **14** are connected to each other via a bus **15**. The CPU **11**, the ROM **12**, and the RAM **13** are configured as a control section **100**. That is, the control section (control operator) **100** performs a process of controlling the code reader **1**, as will be described below, if the CPU **11** operates in accordance with a control program stored in the ROM **12** and/or the memory section **14** and loaded in the RAM **13**.

(27) The RAM **13** includes an image storage section (image storage) **131** and a decoding storage section (decoding storage) **132**. The image storage section **131** stores an image of the commodity T captured by the camera **19**. The decoding storage section **132** stores decoded information (the code information A) read by starting/executing the program A on an image (stored in the image storage section **131** and obtained by decoding the invisible information F). The decoding storage section **132** stores decoded information (the code information B) read by starting/executing the program B on an image (stored in the image storage section **131** and obtained by decoding information regarding the symbol printed on the discount seal R).

(28) The memory section **14** is configured with a nonvolatile memory such as a hard disc drive (HDD) or a flash memory in which stored information is kept although power is turned off. The memory section **14** also includes a control program section (control program storage) **141** that stores a control program for controlling the code reader **1** (a control program causing the code reader **1** to perform control illustrated in FIG. **5**). The control program section **141** stores both the program A and the program B.

(29) The control section **100** is connected to an operation section (input panel) **17**, a display section (display) **18**, and the camera **19** via the bus **15** and a controller **16**.

(30) The operation section **17** is, for example, a keyboard or a touch panel and is operated by an operator of the code reader **1**. The display section **18** is, for example, a liquid crystal display section and displays information for the operator of the code reader **1**.

(31) The control section **100** is connected to a communication section (communication interface) **20** via the bus **15**. The communication section **20** is connected to the POS terminal **3** via the connection line **5** and transmits and receives information to and from the POS terminal **3**.

(32) Next, hardware of the POS terminal **3** will be described. FIG. **4** is a block diagram illustrating an example of a hardware configuration of the POS terminal **3**, according to some embodiments. As illustrated in FIG. **4**, the POS terminal **3** includes a CPU **31**, a ROM **32**, a RAM **33**, and a memory section (memory) **34**. The CPU **31** serves as a control entity. The ROM **32** stores various programs. The RAM **33** loads a program and/or various kinds of data. The memory section **34** stores various programs. The CPU **31**, the ROM **32**, the RAM **33**, and the memory section **34** are connected to each other via a bus **35**. The CPU **31**, the ROM **32**, and the RAM **33** are configured as a control section (control operator) **300**. That is, the control section **300** performs a process of controlling the POS terminal **3**, as will be described below, if the CPU **31** operates in accordance with a control program stored in the ROM **32** or the memory section **34** and loaded in the RAM **33**.

(33) The RAM **33** includes the commodity information section **331** (a storage section, a commodity information storage). The commodity information section **331** stores commodity information regarding the commodity T subjected to the commodity registration process (a commodity code, a commodity name, a commodity price, information indicating reading information from the electronic watermark D, and/or information indicating information read from the discount seal R, and the like).

(34) The memory section **34** is configured with a nonvolatile memory such as a hard disc drive (HDD) or a flash memory in which stored information is kept although power is turned off. The memory section **34** also stores a control program section (control program storage) **341** that stores a control program controlling the POS terminal **3** (a control program causing the POS terminal **3** to perform control illustrated in FIG. **7**) and a commodity master (commodity database) **342**. The commodity master **342** stores commodity information (a commodity name, a commodity price, and the like) regarding the commodity T in association with a commodity code for identifying the

commodity T to be sold in a store.

(35) The control section **300** is connected to an operation section (input panel) **41**, a staff display section (staff display) **42**, a customer display section (customer display) **43**, a card reading section (card reader) **44**, a printer **45**, and a code reader **1** via a bus **35** and a controller **36**.

(36) The operation section **41** is, for example, a keyboard or a touch panel and is operated by an operator of the POS terminal **3**. The staff display section **42** is, for example, a liquid crystal display section and displays information for a staff who is an operator of the POS terminal **3**. The customer display section **43** is, for example, a liquid crystal display section and displays information for a customer. The card reading section **44** reads a customer code for identifying a customer from a card (for example, a credit card, electronic money, or the like) used for settlement. The control section **300** of the POS terminal **3** performs a settlement process related to the customer based on the read customer code. The printer **45** dispenses a receipt or a voucher on which commodity information (a commodity name, a commodity price, or the like) or settlement information regarding the commodity T subjected to the commodity registration process is printed. The code reader **1** is connected to the controller **36** via the connection line **5**.

(37) The control section **300** is connected to a communication section (communication interface) **47** via the bus **35**. The communication section **47** is connected to another POS terminal **3** or a store server (not illustrated) via a communication line and transmits and receives information.

(38) Hereinafter, control of the code reader **1** will be described. FIG. **5** is a flowchart illustrating an example of a control process of the code reader **1**, according to some embodiments. The operator of the code reader **1** performs an operation of holding the commodity T toward the camera **19**. The control section **100** of the code reader **1** determines whether the commodity T (or a portion of commodity T) is captured by the camera **19** (ACT **11**). If the control section **100** determines that a portion of commodity T is not captured (no in ACT **11**), the operator of the code reader **1** may perform another operation (e.g., holding the commodity T closer to the camera **19**, holding the commodity T at a different angle in front of the camera **19**, holding the commodity T in an area with more/less light, etc.). If the control section **100** determines that the commodity T is captured by the camera **19** (Yes in ACT **11**), the control section **100** stores a captured image in the image storage section **131** (ACT **12**).

(39) Subsequently (in response to storing the image), the control section **100** determines whether the started program A reads information regarding the invisible information F from the captured image (ACT **21**). That is, program A determines whether the captured image contains invisible information F. If the control section **100** determines that the information regarding the invisible information F is read (Yes in ACT **21**), then the control section **100** decodes the read information regarding the invisible information F to generate the code information A (ACT **22**). The control section **100** stores the generated code information A in the decoding storage section **132** (ACT **23**). The code information A stored in the decoding storage section **132** includes the commodity code and information indicating that the invisible information F is read. The control section **100** transmits (outputs) the stored code information A to the POS terminal **3** via the connection line **5** (ACT **24**). Then, the control section **100** returns to ACT **11**.

(40) If the program A does not read information regarding the electronic watermark D (No in ACT **21**), the control section **100** determines whether the started program B reads information regarding the visualized symbol printed on the discount seal R from the captured image (ACT **25**). That is, program B determines whether the captured image contains the discount seal R. If the control section **100** determines that the information regarding the symbol printed on the discount seal R is read (Yes in ACT **25**), then the control section **100** decodes the information regarding the symbol to generate the code information B (ACT **26**). The control section **100** stores the generated code information B in the decoding storage section **132** (ACT **27**). The code information B stored in the decoding storage section **132** includes the commodity code, a discount amount, and information indicating that the discount seal is read. The control section **100** transmits (outputs) the stored code

information B to the POS terminal **3** via the connection line **5** (ACT **24**). Then, the control section **100** returns to ACT **11**.
(41) If the program B determines that the information regarding the visualized symbol is not read from the captured image (for example, if the commodity T is not held, the commodity T is not discounted) (No in ACT **25**), the control section **100** returns to ACT **11**.
(42) Hereinafter, a functional configuration of the POS terminal **3** will be described. FIG. **6** is a functional block diagram illustrating an example of a functional configuration of the POS terminal **3**, according to some embodiments. The control section **300** of the POS terminal **3** functions as a storage unit (storage) **301**, a first determination unit (first evaluator) **302**, a second determination unit (second evaluator) **303**, an output unit (output generator) **304**, and/or erasing unit (eraser) **3041** in according with a control program stored in the control program section **341** of the ROM **32** or the memory section **34**.
(43) The storage unit **301** stores the commodity information regarding the commodity T acquired by reading the symbol attached to the commodity T in the commodity information section **331**. Specifically, the storage unit **301** stores the commodity information A subjected to the commodity registration process in the commodity information section **331** based on the code information A received (input) from the code reader **1**. The commodity information A includes the commodity code, the commodity name, and/or the information indicating the reading from the invisible information F. The storage unit **301** stores the commodity information B subjected to the commodity registration process in the commodity information section **331** based on the code information B received (input) from the code reader **1**. The commodity information B includes the commodity code, the commodity name, and/or the information indicating the reading from the discount seal R.
(44) If the discount information attached to the commodity T is read, the first determination unit **302** determines whether the prior commodity information (e.g., the commodity information stored immediately previously by the storage unit **301**) includes the information related to the invisible information F. Specifically, if the code information B is received, the first determination unit **302** determines whether the prior commodity information stored immediately previously in the storage unit **301** includes the information indicating the reading from the invisible information F.
(45) If the first determination unit **302** determines that the information regarding the commodity T stored immediately previously includes the information indicating the reading from the invisible information F, the second determination unit **303** determines whether the commodity code included in the discount information is identical to (matches) the prior commodity code included in the information regarding the commodity T stored in the commodity information section **331**. Specifically, if the prior commodity information A stored immediately previously includes the information indicating the reading from the invisible information F, the second determination unit **303** may determine that the commodity code included in the commodity information B is identical to the commodity code included in the commodity information A.
(46) If the second determination unit **303** determines that the commodity code included in the commodity information B is identical to (matches) the commodity code included in the commodity information A, the output unit **304** does not output the information related to the invisible information F. Specifically, if the second determination unit **303** determines that the commodity code included in the commodity information B is identical to the commodity code included in the commodity information A, the output unit **304** does not output the prior commodity information A stored immediately previously by the storage unit **301** in the commodity information section **331**.
(47) The erasing unit **3041** may be an example of the output unit **304**. If the second determination unit **303** determines that the commodity code included in the commodity information B is identical to the commodity code included in the commodity information A, the erasing unit **3041** erases the prior commodity information stored immediately previously by the storage unit **301** in the commodity information section **331**. Specifically, if the second determination unit **303** determines

that the commodity code included in the commodity information B is identical to the commodity code included in the commodity information A, the erasing unit **3041** erases (that is, does not output) the prior commodity information A stored immediately previously by the storage unit **301** in the commodity information section **331**.

(48) Next, control of the POS terminal **3** will be described. FIG. **7** is a flowchart illustrating an example of a control process of the POS terminal **3**, according to some embodiments. As illustrated in FIG. **7**, the control section **300** of the POS terminal **3** determines whether the code information is received (input) (ACT **31**). The code information is the code information A or the code information B. If the control section **300** determines that the code information is received (input) (Yes in ACT **31**), the control section **300** determines whether the received code information is the code information A (ACT **32**). If the control section **300** determines that the received code information is the code information A (Yes in ACT **32**), then (i) the storage unit **301** reads the commodity information regarding the commodity T identified by the commodity code included in the received code information A from the commodity master **342**; (ii) the control section **300** performs the commodity registration process; and (iii) the control section **300** stores the commodity information Ain the commodity information section **331** (ACT **37**). The control section **300** may store the commodity information in the commodity information section **331** by putting, for example, a mark (for example, an immediately previous flag) so that the commodity information stored immediately previously (the prior commodity information) is distinguished from other commodity information. If subsequent commodity information is stored, the mark is erased and attached to the subsequent commodity information. In this manner, the mark is associated with the most recently stored commodity information (e.g., the immediately previous commodity information). Alternatively, the commodity information stored immediately previously (the prior commodity information) is stored at a special position, distinguishing the prior commodity information with other previous/historic stored commodity information. Then, the control section **300** returns to ACT **31**.

(49) If the control section **300** determines that the received code information is not the code information A (No in ACT **32**), the control section **300** determines whether the code information B related to the discount seal R is received (ACT **33**). If the control section **300** determines that the code information B is received (Yes in ACT **33**), the control section **300** performs the commodity registration process based on the code information (the code information B) related to the discount seal (ACT **34**). In the commodity registration process of ACT **34**, the commodity information is not yet stored in the commodity information section **331**.

(50) The first determination unit **302** determines whether the commodity information stored immediately previously in the commodity information section **331** (the prior commodity information) includes the information indicating the reading from the invisible information F (ACT **35**) (e.g., the commodity information read from the invisible information F of the captured image). If it is determined that the commodity information stored immediately previously in the commodity information section **331** includes the information indicating the reading from the invisible information F (Yes in ACT **35**), the second determination unit **303** determines whether the commodity code included in the commodity information regarding the commodity T subjected to the commodity registration process in ACT **34** (e.g., the commodity information associated with commodity information B) is identical to (matches) the commodity code included in the commodity information A stored immediately previously in the commodity information section **331** (ACT **36**). If it is determined that the commodity codes are not identical to each other (No in ACT **36**), the storage unit **301** stores the commodity information (the commodity information B) regarding the commodity T subjected to the commodity registration process in ACT **34** in the commodity information section **331** (ACT **39**).

(51) On the other hand, if it is determined that the commodity code included in the commodity information regarding the commodity T is identical to the commodity code included in the commodity information A (Yes in ACT **36**), the erasing unit **3041** erases (deletes) the commodity

information stored immediately previously in the commodity information section **331** (ACT **38**). The storage unit **301** stores the commodity information (the commodity information B) regarding the commodity T subjected to the commodity registration process in ACT **34** in the commodity information section **331** (ACT **39**).

(52) If the first determination unit **302** determines that the commodity information stored immediately previously in the commodity information section **331** does not include the information indicating the reading from the invisible information F (No in ACT **35**), the storage unit **301** stores the commodity information (the commodity information B) regarding the commodity T subjected to the commodity registration process in ACT **34** in the commodity information section **331** (ACT **39**). If it is determined that the code information (that is, the code information B) related to the discount seal R is not received (No in ACT **33**), the control section **300** performs another process (ACT **41**). Then, the control section **300** returns to ACT **31**.

(53) If it is determined in ACT **31** that the code information is not received (No in ACT **31**), the control section **300** determines whether a settlement operation is performed (ACT **42**). The settlement operation is, for example, an operation on a settlement button (not illustrated) provided in the operation section **41**. If it is determined that the settlement operation is performed (Yes in ACT **42**), the control section **300** performs the settlement process based on the commodity information stored in the commodity information section **331** (ACT **43**) (e.g., the commodity information of commodity information A and/or commodity information B). The control section **300** ends a transaction with the customer. Conversely, if it is determined that the settlement operation is not performed (No in ACT **42**), the control section **300** returns to ACT **31**.

(54) In this way, the POS terminal **3** according to the first embodiment includes: the commodity information section **331**; the storage unit **301** configured to store the commodity information regarding the commodity T acquired by reading the symbol attached to the commodity T in the commodity information section **331**; the first determination unit **302** configured to determine (if discount information attached to the commodity T is read) whether the commodity information stored immediately previously by the storage unit **301** (e.g., the prior commodity information) is the commodity information A including information related to the invisible information F; the second determination unit **303** configured to determine (if the first determination unit **302** determines that the commodity information stored immediately previously includes the information indicating the reading from the invisible information F) whether the commodity code included in the commodity information B is identical to the commodity code included in the commodity information A stored in the commodity information section **331**; and the erasing unit **3041** configured to erase (if the second determination unit **303** determines that the commodity code included in the commodity information B is identical to commodity code included in the commodity information A) the information regarding the commodity T stored immediately previously in the storage unit **301** from the commodity information section **331**.

(55) If the second determination unit **303** determines that both the commodity codes (the commodity code associated with commodity information A and the commodity code associated with commodity information B) are identical to each other, the erasing unit **3041** in the POS terminal **3**, according to the first embodiment, erases the information regarding the commodity T stored immediately previously by the storage unit **301** in the commodity information section **331** (e.g., the prior commodity code associated with commodity information A). Accordingly, double registration due to the invisible information F and the discount seal may be prevented.

Second Embodiment

(56) A second embodiment will be described below. In the second embodiment, the code reader **1** will be described as an example of the information processing device. It should be appreciated that the second embodiment is an example embodiment of the information processing device, and the method, configurations, functions, and the like do not limit the example embodiment. Some descriptions of configurations in the second embodiment may be similar to those of the first

embodiment (the configurations of FIGS. **1** to **3**). Accordingly, the configurations of FIGS. **1** to **3** will be omitted or simplified. For example, FIG. **3** may also illustrate a hardware configuration of the code reader **1** according to the second embodiment. The code reader **1** according to the second embodiment differs from that of the first embodiment in the control program stored in the control program section **141**. In addition to the programs A and B, a control program causing the code reader **1** to perform control illustrated in FIG. **9** is stored.
(57) Next, a functional configuration of the code reader **1** according to the second embodiment will be described. FIG. **8** is a functional block diagram illustrating an example of a functional configuration of the code reader **1** according to the second embodiment. The control section (control operator) **100** of the code reader **1** functions as the storage unit (storage) **101**, the first determination unit (first evaluator) **102**, the second determination unit (second evaluator) **103**, the output unit (output generator) **104**, and/or the erasing unit (eraser) **1041** in accordance with the program A, the program B, and the control program stored in the control program section **141** of the memory section **14** and/or the ROM **12**.
(58) The storage unit **101** stores code information of the commodity T acquired by reading a symbol attached to the commodity T in the decoding storage section **132** (a storage section, decoding storage). Specifically, the storage unit **101** stores decoded code information A obtained by executing the program A and decoding the invisible information F read from the electronic watermark D in the decoding storage section **132** among other image information captured by the camera **19** and stored in the image storage section **131**. The code information A includes a commodity code and/or information indicating the reading from the invisible information F. The storage unit **101** stores decoded code information B obtained by starting the program B and decoding the symbol read from the discount seal R in the decoding storage section **132** among image information captured by the camera **19** and stored in the image storage section **131**. The code information B includes a commodity code, a commodity price, a discount amount, and/or information indicating the reading from the discount seal R.
(59) If the discount information attached to the commodity T is read, then the first determination unit **102** determines whether the information regarding the commodity T stored immediately previously by the storage unit **101** (the prior code information A) includes the information related to the invisible information F. Specifically, if the storage unit **101** stores the code information B in the decoding storage section **132**, then the first determination unit **102** determines whether the code information stored immediately previously in the storage unit **101** (the prior code information A) includes the information indicating the reading from the invisible information F.
(60) If the first determination unit **102** determines that the information regarding the commodity T stored immediately previously includes the information indicating that the invisible information F is read, then the second determination unit **103** determines whether the commodity code included in the discount information is identical to (matches) the commodity code included in the information regarding the commodity T stored in the decoding storage section **132**. Specifically, if it is determined that the information regarding the commodity T stored immediately previously includes the information indicating that the invisible information F is read, the second determination unit **103** determines whether the commodity code included in the code information B (the discount information) is identical to the commodity code included in the code information A.
(61) If the second determination unit **103** determines that the commodity code included in the code information B is identical to the commodity code included in the code information A, the output unit **104** does not output the information related to the invisible information F. Specifically, if the second determination unit **103** determines that the commodity code included in the code information B is identical to the commodity code included in the code information A, the output unit **104** does not output the code information A stored immediately previously by the storage unit **101** in the decoding storage section **132** (e.g., the prior code information A).
(62) The erasing unit **1041** is an example of the output unit **104**. If the second determination unit

103 determines that the commodity code included in the code information B is identical to the commodity code included in the code information A, then the erasing unit **1041** erases the code information stored immediately previously by the storage unit **101** in the decoding storage section **132**. Specifically, if the second determination unit **103** determines that the commodity code included in the code information B is identical to the commodity code included in the code information A, the erasing unit **1041** erases (that is, does not output) the code information A stored immediately previously in the decoding storage section **132** by the storage unit **101**.
(63) If the second determination unit **103** determines that the commodity code included in the code information B is not identical to the commodity code included in the code information A, the erasing unit **1041** outputs the decoded information decoded immediately previously and the decoded discount information to the POS terminal **3** connected to the code reader **1**. If the second determination unit **103** determines that the commodity code included in the code information B is identical to the commodity code included in the code information A, the decoded discount information is output to the POS terminal **3**. Specifically, if the second determination unit **103** determines that the commodity code included in the code information B is not identical to the commodity code included in the code information A, the erasing unit **1041** outputs the decoded information (the prior code information A) decoded immediately previously and stored in the decoding storage section **132** and the decoded discount information (the code information B) to the POS terminal **3** connected to the code reader **1**. If the second determination unit **103** determines that the commodity code included in the code information B is identical to the commodity code included in the code information A, the erasing unit **1041** outputs the decoded discount information (the code information B) to the POS terminal **3**.
(64) Next, control of the code reader **1** will be described. FIG. **9** is a flowchart illustrating an example of a control process of the code reader **1** according to the second embodiment. As illustrated in FIG. **9**, the control section **100** of the code reader **1** captures the commodity T (ACT **11**) and stores the captured image in the image storage section **131** (ACT **12**). Subsequently, the control section **100** determines whether the started program A reads the information regarding the invisible information F from the captured image (ACT **51**). If the control section **100** determines that the information regarding the invisible information F is read (Yes in ACT **51**), the control section **100** decodes the read information regarding the invisible information F to generate the code information A (ACT **52**). The code information A includes the commodity code and/or information indicating reading of the invisible information F (the information related to the invisible information).
(65) Subsequently, the control section **100** determines whether the code information A is stored in the decoding storage section **132**. If the code information A is stored, the code information A is transmitted to the POS terminal **3** (ACT **53**). If the code information A is not stored in the decoding storage section **132**, the process of ACT **53** is not performed. Subsequently, the storage unit **101** stores the code information A generated in ACT **52** in the decoding storage section **132** (ACT **54**). The code information A stored in the decoding storage section **132** includes the commodity code and information indicating reading of the invisible information F (the information related to the invisible information). That is, the decoding storage section **132** stores the immediately previous code information A (the prior code information A). Then, the control section **100** returns to ACT **11**.
(66) If it is determined that the information regarding the invisible information F is not read (No in ACT **51**), the control section **100** determines whether the started program B reads the information regarding the symbol printed on the discount seal R from the captured image (ACT **61**). If the control section **100** determines that the information regarding the symbol printed on the discount seal R is read (Yes in ACT **61**), the control section **100** decodes the information regarding the symbol to generate the code information B (ACT **62**). The code information B includes the commodity code, the discount amount, and/or the information indicating that the discount seal is read.

(67) Subsequently, the first determination unit **102** determines whether the code information stored immediately previously in the decoding storage section **132** (the prior code information A) includes the information indicating the reading from the invisible information F (ACT **63**). If it is determined that the code information stored immediately previously in the decoding storage section **132** includes the information indicating the reading from the invisible information F (Yes in ACT **63**), the second determination unit **103** determines whether the commodity code included in the code information B decoded in ACT **62** is identical to the commodity code included in the code information A stored immediately previously in the decoding storage section **132** (ACT **64**). If it is determined that the commodity code included in the code information B is not identical to the commodity code included in the code information A (No in ACT **64**), the storage unit **101** stores the code information (the commodity information B) of the commodity T decoded in ACT **62** in the decoding storage section **132** (ACT **67**). The output unit **105** transmits (outputs) all the code information (the code information A stored in ACT **54** and the code information B stored in ACT **67**) to be stored in the decoding storage section **132** to the POS terminal **3** (ACT **68**). Then, the control section **100** returns to ACT **11**.
(68) If it is determined that the commodity code included in the code information B is identical to the commodity code included in the code information A (Yes in ACT **64**), the erasing unit **1041** erases the code information A stored in ACT **54** from the decoding storage section **132** (ACT **65**). The storage unit **101** stores the code information (the commodity information B) of the commodity T decoded in ACT **62** in the decoding storage section **132** (ACT **67**). The output unit **105** transmits (outputs) the code information (the code information B stored in ACT **67**) stored in the decoding storage section **132** to the POS terminal **3** (ACT **68**). Then, the control section **100** returns to ACT **11**.
(69) If it is determined that the code information stored immediately previously in the decoding storage section **132** does not include the information indicating the reading from the invisible information F (No in ACT **63**), the storage unit **101** stores the code information (the commodity information B) of the commodity T decoded in ACT **62** in the decoding storage section **132** (ACT **67**). The output unit **105** transmits (outputs) the code information (the code information B stored in ACT **67**) stored in the decoding storage section **132** to the POS terminal **3** (ACT **68**). Then, the control section **100** returns to ACT **11**.
(70) In this way, the code reader **1** according to the second embodiment includes: the decoding storage section **132**; the storage unit **101** configured to store the information regarding the commodity T acquired by reading the symbol attached to the commodity T in the decoding storage section **132**; the first determination unit **102** configured to determine (if discount information attached to the commodity T is read) whether the commodity T information (the code information A) stored immediately previously by the storage unit **101** includes information related to the invisible information F; the second determination unit **103** configured to determine (if the first determination unit **102** determines that the information regarding the commodity T stored immediately previously includes the information related to the invisible information F) whether the commodity code included in the discount information is identical to commodity code included in the code information A; and the erasing unit **1041** configured to erase (if the second determination unit **103** determines that the commodity code included in the discount information is identical to the commodity code included in the code information A) the code information A stored immediately previously in the storage unit **101** from the decoding storage section **132**.
(71) If the second determination unit **103** determines that both the commodity codes (the commodity code associated with code information A and the commodity code associated with code information B) are identical to each other, the code reader **1**, according to the second embodiment erases the information regarding the commodity T stored immediately previously by the storage unit **101** in the decoding storage section **132**. Accordingly, double registration due to the invisible information F and the discount seal may be prevented.

(72) While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the inventions. Indeed, the novel embodiments described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions and changes in the form of the embodiments described herein may be made without departing from the spirit of the inventions. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the inventions.
(73) For example, in the first and second embodiments, a case where the commodity code is included in the invisible information F is described. However, the example embodiment is not limited thereto and the commodity code stored in advance in another memory provided in the code reader **1** or the POS terminal **3** may be acquired based on the invisible information F.
(74) In the first and second embodiments, the electronic watermark D is printed on a commodity. However, the example embodiment is not limited thereto. For example, labels on which the electronic watermark D is printed may be attached to a plurality of surfaces of a commodity.
(75) In the first and second embodiments, the erasing units (**3041** and **1041**) are used as an example of the output units (**304** and **104**) in the description. However, the example embodiment is not limited thereto. The output units (**304** and **104**) may be other units other than the erasing units (**3041** and **1041**) as long as the commodity information stored immediately previously (prior commodity information) or decoded information (prior code information) is not output.
(76) The programs executed in the code reader **1** or the POS terminal **3** according to the first and second embodiments are recorded and provided as files with an installable format or an executable format on a computer-readable recording medium such as a CD-ROM, a flexible disc (FD), a CD-R, or a digital versatile disk (DVD).
(77) The programs executed in the code reader **1** and the POS terminal **3** according to the first and second embodiments may be stored on a computer connected to a network such as the Internet and may be downloaded via the network to be provided. The programs executed in the code reader **1** and the POS terminal **3** according to the first and second embodiments may be provided or distributed via a network such as the Internet.
(78) The programs executed in the code reader **1** and the POS terminal **3** according to the first and second embodiments may be each embedded in a ROM or the like to be provided.

Claims

1. An information processing device comprising: a camera configured to read a symbol attached to a commodity and capture an image containing discount information attached to the commodity, and read invisible information printed on one or more surfaces of the commodity, wherein the invisible information is invisible to a human eye; a storage storing information regarding the commodity acquired by reading the symbol attached to the commodity; and one or more processors configured to: decode the invisible information; determine whether prior commodity information stored in the storage includes information related to the invisible information, in response to reading discount information attached to the commodity; determine whether commodity identification information included in the discount information is identical to commodity identification information included in the information related to the invisible information, in response to determining that the prior commodity information stored in the storage includes the information related to the invisible information; and output information related to the discount information, in response to determining that the commodity identification information included in the discount information is identical to the commodity identification information included in the information related to the invisible information.
2. The information processing device according to claim 1, wherein the information processing device is a commodity sales data processing device.

3. The information processing device according to claim 1, wherein the camera is a code reader configured to execute: a program A configured to read the symbol related to the invisible information; and a program B configured to read the discount information attached to the commodity.
4. The information processing device according to claim 1, wherein the storage stores at least the storage, the first storage storing commodity information determined via a commodity registration process based on information read by the camera regarding the commodity.
5. The information processing device according to claim 1, wherein the prior commodity information is commodity information stored immediately previously by the storage.
6. The information processing device according to claim 1, wherein the the one or more processors are configured to erase the prior commodity information, in response to determining that the commodity identification information included in the discount information is identical to the commodity identification information included in the commodity information.
7. The information processing device according to claim 1, wherein: the storage stores decoded information obtained by decoding the image captured by the camera, the stored decoded information including prior decoded information stored immediately previously by the storage.
8. The information processing device according to claim 7, wherein the one or more processors are configured to determine whether the prior decoded information stored by the storage includes the information related to the invisible information, in response to the captured image containing the discount information attached to the commodity.
9. The information processing device according to claim 7, wherein the one or more processors are configured to determine whether the commodity identification information included in the discount information is identical to commodity identification information included in the prior decoded information.
10. The information processing device according to claim 7, wherein the one or more processors are configured to erase the prior decoded information, in response to determining that the commodity identification information included in the discount information is identical to commodity identification information included in the decoded information.
11. The information processing device according to claim 7, wherein the one or more processors are configured to output the decoded information and decoded discount information to a commodity sales data processing device connected to the information processing device, in response to determining that the commodity identification information included in the discount information is not identical to commodity identification information included in the decoded information.
12. The information processing device according to claim 7, wherein the one or more processors are configured to output the decoded discount information to a commodity sales data processing device connected to the information processing device, in response to determining that the commodity identification information included in the discount information is identical to commodity identification information included in the decoded information.
13. A method for preventing double registration comprising: reading a symbol attached to a commodity or discount information attached to the commodity, and reading invisible information printed on one or more surfaces of the commodity, wherein the invisible information is invisible to a human eye; storing information regarding the commodity acquired by reading the symbol; decoding, by the one or more processors, the invisible information; responsive to reading discount information, determining, by one or more processors, whether stored prior commodity information includes information related to invisible information; responsive to determining that the prior commodity information includes the information related to invisible information, determining, by the one or more processors, whether commodity identification information included in the discount information is identical to commodity identification information included in the information related to the invisible information; and responsive to determining that the commodity identification

information included in the discount information is identical to the commodity identification information included in the information related to the invisible information, outputting information related to the discount information.

14. The method of claim 13, wherein the prior commodity information is commodity information stored immediately previously by a storage.

15. The method of claim 13, further comprising: responsive to determining that the commodity identification information included in the discount information is identical to the commodity identification information included in the commodity information, erasing, the prior commodity information.

16. The method of claim 13, further comprising: responsive to determining that a captured image contains discount information attached to the commodity, determining, by the one or more processors, whether stored prior decoded information includes the information related to the invisible information.

17. The method of claim 13, further comprising: determining, by the one or more processors, whether the commodity information included in the discount information is identical to commodity information included in stored prior decoded information.

18. The method of claim 13, further comprising: erasing stored prior decoded information in response to determining that the commodity identification information included in the discount information is identical to commodity identification information included in decoded information obtained by decoding a captured image.

19. The method of claim 13, further comprising: responsive to determining that the commodity identification information included in the discount information is not identical to commodity identification information included in decoded information obtained by decoding a captured image, outputting the decoded information and decoded discount information to a commodity sales device.

20. The method of claim 13, further comprising: responsive to determining that the commodity identification information included in the discount information is identical to commodity identification information included in decoded information obtained by decoding a captured image, outputting the decoded discount information to a commodity sales device.
